Tanay Agrawal

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Education

University of Southern California (USC) — Los Angeles, CA
M.S. Electrical Engineering | Research Assistant, WiDeS Group | GPA: 3.75/4 — *Aug 2025 – Aug 2027*
- Coursework: Wireless Communications, Machine Learning, Linear Algebra, Probability

BITS Pilani K K Birla Goa Campus — Goa, India
B.E. Electronics and Communication Engineering | GPA: 8.32/10 — *Aug 2021 – Aug 2025*
- Bachelor's Thesis: *Linear Channel Modeling and Linear Detectors for Multi-Mode Fibers* | TU Munich

Skills

Lab Instruments: VNA, Anechoic Chamber, AWG, Oscilloscopes, Signal Analyzers, Phased Arrays, GPSDO, IF-over-Fiber Systems
Programming: Python, MATLAB, C/C++
Software: Simulink, STM32 Cube IDE, ESP-IDF, DWM-SDK CST Studio Suite, GNU Radio, Vivado (Model-based design)
Protocols: 802.11 (WiFi), BLE (NimBLE), SPI, UART, I2C
Prototyping Skills: PCB design in KiCAD/OrCAD, Soldering, Circuit design

Experience

USC Viterbi | WiDeS Group — Los Angeles, CA
Research Assistant under Prof. Andreas Molisch — *Aug 2025 – Present*
- **RFSoC FMCW Radar (DARPA funded):** Designed and programmed a complete channel sounder at 6.5 GHz using Zynq UltraScale+ RFSoC, GPSDO-disciplined timing, PA, LNA, BPF, including RF chain characterization, timing and stability testing, waveform generation. Installed a custom MIMO antenna array for FMCW Radar Application.
- **Distributed MIMO Channel Sounder Infrastructure:** Integrated GPSDO-disciplined timing, 2-stage LO chain, IF-over-fiber interconnects, and PPS/10 MHz reference distribution across three transmitter carts and a receiver cart for extensive outdoor urban mmWave measurements. Contributed to STM32 MCU firmware and full array calibration in anechoic chamber.
- **PDP MPC Extraction for Cell-Free Massive MIMO:** Designed multi-stage MATLAB pipeline for 3–4 GHz channel data using 2D image-based peak detection, DBSCAN clustering, and constrained least-squares solver for multipath boundary identification to extract PDP components for measurement data.

BITS Pilani | Student Researcher — Goa, India
Research Assistant under Prof. Ravi Kadlimatti — *Jan 2024 – Aug 2025*
- **Code-Aided Delay-Doppler Domain Channel Estimation:** Novel contribution in using CAZAC sequences in embedded pilot aided delay-Doppler domain channel estimation for OTFS *Published: IEEE CCNC 2026.*
- **Equalization Techniques for OTFS in UWA Environments:** Conducted a study of OTFS in realistic UWA oceanic environment channel scenarios with high doppler and compared equalization methods such as ZF, LMMSE, MRC, MPA. *Published: APWiMob 2025.*

Projects

ESP32 BLE Scanner with Anechoic Chamber Validation | *C++, Python, Matlab* — Jun 2026
- Built dual-core BLE scanner using NimBLE radio stack on ESP32 WROOM 32E. Validated MATLAB-based RSSI measurements with Python CSV logging pipeline against theoretical path loss model in anechoic chamber.

Drone RF Signal Classification | *Python, NumPy, scikit-learn* — March 2026 – May 2026
- Engineered 89-feature IQ signal pipeline (time-domain, spectral, phase, burst statistics) from 17,000+ raw captures with hierarchical SVM + MLP ensemble achieved 66.9% balanced accuracy across −12 to +30 dB SNR.

Scalable Ku/K/C-Band Microstrip Patch Antenna Array | *CST Studio Suite* — Apr 2024 – Jul 2024
- Designed scalable RT-duroid 5880LZ patch array with coaxial probe feed achieving consistent 3 dBi directivity increase per element doubling (11.47 to 26.36 dBi, 1 to 32 elements); validated across Ku (13.97 GHz), K (22.66 GHz), and C (7.6 GHz) bands via linear dimension scaling. *Submitted: PIER Journal.*

Publications

[1] Agrawal, T., & Kadlimatti, R. (2026). "Embedded pilot code-aided delay-Doppler domain channel estimation." *IEEE CCNC 2026*, Las Vegas, NV.

[2] Agrawal, T., & Kadlimatti, R. (2025). "Comparison of channel equalization methods for an OTFS-based underwater acoustic communication system." *APWiMob 2025.*

[3] Agrawal, T., & Baudha, S. "A Scalable Microstrip Patch Antenna Array for Satellite Applications." *Under review, PIER Journal.*